

Week 8 Profiling Report

System Performance, Stress Burst and Telemetry Analysis

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Project: Phase 2 Backend Development Project

Week: 8

1. Objective

Evaluate system behavior under a simulated high-frequency event burst using the asynchronous worker queue, live log processing, Prometheus telemetry, and responsive dashboard.

2. Test Configuration

100 stress_burst events were generated. Events were processed asynchronously and written to logs/application.log. Prometheus-compatible telemetry was collected using prom-client, with the dashboard refreshing every 5 seconds.

3. Stress Test Results

Metric	Result
Events generated	100
Events recorded	100
Average simulated latency	504.33 ms
Peak simulated latency	987 ms
Total processing/test duration	3.015 seconds

4. Analysis

All 100 generated events were successfully recorded. The average simulated latency was 504.33 ms and the peak was 987 ms. The worker processed the burst asynchronously without placing queue processing inside database transaction blocks.

5. Telemetry

The Prometheus endpoint exposes HTTP request count and duration, worker events processed, worker processing duration, queue size, stress-test latency, and Node.js CPU, memory, event-loop, and garbage-collection metrics.

6. Verification

Prometheus /metrics: PASS
Worker queue processing: PASS
Live JSON log recording: PASS
100-event stress burst: PASS
Responsive telemetry dashboard: PASS
Peak latency measurement: PASS

7. Conclusion

The Week 8 stress-burst profiling test completed successfully. All 100 generated events were recorded and performance telemetry was exposed through Prometheus-compatible metrics and the responsive dashboard. The peak simulated latency for this run was 987 ms.